A Project Report on

Fusion Based Video Summarization Integrating Transcripts and Key Frames for YouTube Content Analysis

Submitted in partial fulfilment for award of
Bachelor of Technology

Degree

in

CSE (Artificial Intelligence and Machine Learning)

By

N. Harsha Vardhan(Y22ACM455) **Sk. Reena Kowsar(Y22ACM470)**

Sk. Neha(Y22ACM469) **P. Pavan Naga Sai(Y22ACM461)**

Under the guidance of
Vasireddy Swathi, M.tech
Assistant Professor

Department of CSE (Artificial Intelligence and Machine Learning)
Bapatla Engineering College
(Autonomous)
(Affiliated to Acharya Nagarjuna University)
BAPATLA – 522 102, Andhra Pradesh, INDIA

2025-2026

Department of

CSE (Artificial Intelligence and Machine Learning)

CERTIFICATE

This is to certify that the project report entitled **Fusion Based Video Summarization Integrating Transcripts and Key Frames for YouTube Content Analysis** being submitted by NADENDLA HARSHA VARDHAN(Y22ACM455),SHAIK REENA KOWSAR(Y22ACM470), SHAIK NEHA(Y22ACM469),PAMUJULA PAVAN NAGA SAI(Y22ACM461) in partial fulfillment for the award of the Degree of Bachelor of Technology in CSE (Artificial Intelligence and Machine Learning) to Acharya Nagarjuna University is a record of bonafide work carried out by them under our guidance and supervision.

Date:

Signature of the Guide	**Signature of the HOD**
Vasireddy Swathi	**Dr. K. Manideep**
Assistant Professor	**Assoc. Prof. & Head**

DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

Nadendla Harsha Vardhan (Y22ACM455)

Shaik Reena Kowsar (Y22ACM470)

Shaik Neha (Y22ACM469)

Pamujula Pavan Naga Sai (Y22ACM461)

Acknowledgement

We sincerely thank the following distinguished personalities who have given their advice and support for successful completion of the work.

We are deeply indebted to our most respected guide **Ms. Vasireddy Swathi,** Assistant Professor, Department of CSE(AIML), for her valuable and inspiring guidance, comments, suggestions and encouragement.

We extend our sincere thanks to **Dr. K. Manideep**, Assoc. Prof. & Head, Department of CSE(AIML) for extending his cooperation and providing the required resources.

We would like to thank our beloved Principal **Dr. B. Srinivasa Rao** for providing the online resources and other facilities to carry out this work.

We would like to sincerely thank our project coordinator **Ms. G. V. Leela Kumari**, Assistant Prof., Dept. of CSE(AIML), for her helpful suggestions in presenting this document.

We extend our sincere thanks to all other teaching faculty and non-teaching staff of the department, who helped directly or indirectly for their coopertion and encouragement.

Nadendla Harsha Vardhan (Y22ACM455)

Shaik Reena Kowsar (Y22ACM470)

Shaik Neha (Y22ACM469)

Pamujula Pavan Naga Sai (Y22ACM461)

ABSTRACT

The rapid growth of online video platforms such as YouTube has resulted in an enormous volume of video content, creating significant challenges for users seeking to efficiently extract important information from lengthy videos. Traditional video summarization methods rely primarily on either visual features or textual content, often failing to produce comprehensive and informative summaries. This project proposes a Fusion Based Video Summarization System that integrates multiple sources of information including video transcripts, keyframes, and engagement metrics to generate intelligent summaries of YouTube videos. The system extracts transcripts using the YouTube Transcript API and applies Natural Language Processing techniques including TextRank and TF-IDF algorithms to identify the most significant sentences. An engagement-aware fusion technique is incorporated which considers viewer retention patterns, interaction peaks, and sentiment variations to highlight the most important moments. The system further employs the Gemini Generative AI model to produce refined abstractive summaries with improved readability. The final output is presented through a Streamlit-based web interface where users can input a YouTube video URL and receive structured summaries with clickable timestamps and keyframe highlights.

Keywords: Video Summarization, TextRank, TF-IDF, Engagement Fusion, Keyframe Extraction, Generative AI, YouTube Content Analysis, Natural Language Processing

TABLE OF CONTENTS

ABSTRACT v
TABLE OF CONTENTS vi
LIST OF TABLES viii
LIST OF FIGURES ix
1. Introduction 1
1.1 Background of the Study 1
1.2 Problem Statement 2
1.3 Objectives 3
1.4 Scope 4
1.5 Organisation of the Report 5
2. Literature Review 6
2.1 Overview of Existing Work 6
2.2 Comparison of Previous Approaches 8
3. System Analysis 9
3.1 Existing System 9
3.2 Proposed System 10
3.3 Feasibility Study 11
3.4 Requirements 12
3.4.1 Hardware Requirements 12
3.4.2 Software Requirements 12
3.4.3 Functional Requirements 13
3.4.4 Non-Functional Requirements 13
4. Software Design 15
4.1 Architecture Design 15
4.2 UML Diagrams 17
4.2.1 Class Diagram 17
4.2.2 Interaction Diagrams 19
4.2.2.1 Sequence Diagram 19
4.2.2.2 Collaboration Diagram 21
4.2.3 Object Diagram 22
4.2.4 Use Case Diagram 23
4.2.5 Activity Diagram 25
4.2.6 Control Flow Diagram 27
4.3 Database Design 29
4.4 Algorithm / Design Methodology 30
5. Implementation 34
5.1 Technologies Used 34
5.2 Modules Description 35
5.2.1 Task Management Component 35
5.2.2 Predictive Intelligence Component 35
5.2.3 Adaptive Scheduling Component 36
5.2.4 Wellbeing Optimization Component 36

5.2.5 Generative AI Explanation Component 36
5.2.6 Closed-Loop Feedback Learning Mechanism 36
5.2.7 System Support and Interaction Component 37
5.3 Coding 37
5.3.1 Project Folder Structure 37
5.3.2 Backend Implementation 37
5.3.3 Frontend Implementation 39
6. Testing 42
6.1 Test Cases 44
6.2 Testing Methods 44
6.2.1 Unit Testing 45
6.2.2 Integration Testing 45
6.2.3 System Testing 46
6.3 Results and Analysis 46
7. Results and Discussion 47
7.1 Dashboard Overview 48
7.2 Video Input Interface 48
7.3 AI-Based Summary Generation 49
7.4 Key Frame / Highlight Visualization 49
7.5 Summary Analytics and Insights 49
7.6 Explainable AI-Based Output 50
7.7 System Overall Performance 50
8. Conclusion and Future Work 52
8.1 Summary of the Project 52
8.2 Achievements 52
8.3 Limitations 52
8.4 Future Enhancements 53
9. References 54

LIST OF TABLES

Table No.	Table Name	Page No.
Table 2.1	Comparison of Existing Video Summarization Approches	7
Table 6.1	Test Cases of Video Summarization System	44
Table 7.7	Comparsion of summarization technique	51

LIST OF FIGURES

Figure No.	Figure Name	Page No.
Fig 4.1	Fusion-Based Video Summarization System Architecture	16
Fig 4.2.1	Class Diagram	19
Fig 4.2.2.1	Sequence Diagram	20
Fig 4.2.2.2	Collaboration Diagram	22
Fig 4.2.3	Object Diagram	23
Fig 4.2.4	Use Case Diagram	25
Fig 4.2.5	Activity Diagram	27
Fig 4.2.6	Control Flow Diagram	28
Fig 7.1	Dashboard Interface of the Fusion based video summarization	47
Fig 7.3	AI summary of the Fusion based video summarization	48
Fig 7.4	TextRank Summary With Engagement Fusion	49
Fig 7.5	Engagements spikes	50

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

In recent years, the rapid growth of online video platforms such as YouTube has led to an enormous increase in multimedia content. Millions of hours of video are uploaded daily, covering areas such as education, entertainment, news, and tutorials. While this abundance of content provides valuable information, it also creates a challenge for users who want to quickly extract key insights without watching entire videos.

Traditional video summarization techniques primarily rely on either visual features (such as frames and scenes) or textual data (such as transcripts). However, these approaches often lack accuracy and contextual understanding when used independently. Visual-only methods may miss important spoken information, while text-based methods may ignore emotional tone, emphasis, and engagement cues present in audio and video.

With the advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), more sophisticated approaches have emerged that combine multiple data sources. In particular, transcript-based analysis has gained importance, as it allows systems to understand the semantic meaning of spoken content. Additionally, audio and visual signals can provide insights into user engagement, sentiment, and important highlights within a video.

This project focuses on developing a Fusion-Based Video Summarization System that integrates transcripts with audio and visual features to generate accurate and meaningful summaries of YouTube videos. The system uses a hybrid approach that combines extractive summarization (selecting key sentences) and generative techniques (producing human-like summaries), along with engagement analysis to identify the most relevant parts of the video.

By leveraging multimodal data fusion, the proposed system aims to overcome the limitations of existing methods and provide concise, context-aware summaries. This

approach is particularly useful for students, researchers, and professionals who need to quickly understand video content without investing significant time.

Overall, this study highlights the importance of integrating multiple data sources in video analysis and demonstrates how AI-driven techniques can enhance the efficiency and effectiveness of content consumption.

1.2 PROBLEM STATEMENT

With the exponential growth of video content on platforms such as YouTube, users are often overwhelmed by the large volume and length of videos available. Extracting relevant information from these videos requires significant time and effort, especially for educational, technical, or informational content.

Existing video summarization techniques face several limitations. Traditional methods that rely only on visual features fail to capture the actual meaning conveyed through speech. Similarly, text-based approaches that use transcripts alone may overlook important contextual cues such as tone, emphasis, and viewer engagement. As a result, the generated summaries are often incomplete, less accurate, or not contextually meaningful.

Moreover, many current systems do not effectively combine multiple modalities (text, audio, and visual data) to produce a comprehensive summary. The lack of integration leads to poor performance in identifying key highlights and delivering high-quality summaries. Additionally, most available tools provide generic summaries without allowing customization based on user preferences such as summary length, tone, or type.

Therefore, there is a need for an intelligent system that can efficiently analyze YouTube videos by integrating transcripts with audio and visual features, and generate accurate, concise, and context-aware summaries. The system should also be capable of combining extractive and generative approaches to improve the quality of the summary while adapting to user requirements.

This project aims to address these challenges by developing a Fusion-Based Video Summarization System that leverages multimodal data and advanced AI techniques to enhance the effectiveness of video content analysis.

1.3 OBJECTIVES

- To develop an automated system that can summarize long YouTube videos efficiently.
- To extract transcripts from YouTube videos and process them using Natural Language Processing techniques.
- To apply Text Rank and TF-IDF algorithms for identifying important sentences in the transcript.
- To generate timestamp-based keyframes that highlight important segments of the video.
- To combine transcript analysis with engagement-based metrics such as viewer retention and sentiment patterns.
- To implement a fusion-based summarization approach that integrates textual and engagement data.
- To design a user-friendly web interface that allows users to input a video link and view summarized results.
- To provide clickable timestamps and structured notes for easy navigation through the video content.

1.4 SCOPE

The scope of the proposed Fusion-Based Video Summarization Integrating Transcript and Engagement Signals for YouTube Content Analysis focuses on developing an intelligent and scalable system for automatically summarizing long-form video content using a combination of transcript analysis, viewer engagement data, and artificial intelligence techniques.

The system is designed to process YouTube videos that provide transcript information and associated engagement signals, and generate concise, meaningful summaries that capture both the semantic importance and audience impact of the content. Unlike traditional approaches that rely only on textual features, the proposed system integrates

behavioral insights such as retention patterns, interaction peaks, and sentiment intensity to improve the relevance and quality of summaries.

The scope of the project includes the following:

- Extraction of transcripts and metadata from YouTube videos using transcript retrieval mechanisms
- Preprocessing of textual data using Natural Language Processing techniques such as tokenization, normalization, and stop-word removal
- Feature extraction using TF-IDF to represent sentence importance and relationships
- Generation of extractive summaries using graph-based ranking techniques such as the TextRank algorithm
- Analysis of viewer engagement signals including retention rates, interaction peaks, and sentiment intensity
- Temporal alignment of engagement data with transcript segments for identifying high-impact video portions
- Integration of textual and engagement features using a weighted fusion model to compute unified importance scores
- Optimization of engagement features using dimensionality reduction techniques such as Principal Component Analysis
- Enhancement of summaries using generative artificial intelligence for improved readability and coherence
- Presentation of summarized output, key highlights, and engagement insights through an interactive web-based user interface

The system is capable of handling videos from multiple domains such as educational lectures, tutorials, and informational content. It enables users to efficiently understand key ideas without watching entire videos, thereby reducing time consumption and improving content accessibility.

While the current system focuses primarily on transcript and engagement-based analysis, visual features such as key frame extraction are limited and can be further enhanced in future work. The proposed framework can be extended to support

multimodal summarization, multilingual processing, real-time video analysis, and adaptive personalization based on user preferences.

1.5 ORGANISATION OF THE REPORT

This report is organized into multiple chapters for systematic presentation of the proposed Fusion-Based Video Summarization Integrating Transcripts and Key Frames for YouTube Content Analysis. Chapter 1 introduces the project including background, problem statement, objectives, and scope. Chapter 2 presents the literature review and comparison of existing video summarization approaches along with their limitations. Chapter 3 discusses system analysis including the existing system, proposed system, feasibility study, and system requirements. Chapter 4 explains system design including architecture, UML diagrams, database design, and methodology used. Chapter 5 describes implementation details, modules, and technologies used. Chapter 6 presents testing strategies, test cases, and results analysis. Chapter 7 illustrates system outputs, performance evaluation, and comparison with existing methods. Finally, Chapter 8 provides the conclusion along with future enhancements, followed by references listing all sources used.

CHAPTER 2

LITERATURE REVIEW

2.1 OVERVIEW OF EXISTING WORK

In graph based ranking techniques like TextRank, similarities among textual units similar to a graph are constructed to find out important sentences in the input text by applying iterative scoring mechanisms. These methods do not require any labeled data and directly work on longer transcripts efficiently so they are used for extractive summarization. Yet they mainly focus on text relevance, but do not take into account viewer engagement signals. [1]

Thirdly Transformer Based Summarization Models: With the rapid progress in natural language processing over the past recent years, transformer based architectures like BERT or GPT can be used for generating abstractive summarizations by learning context-aware representations of text. While these models yield fluent summaries, they can often be unexplainable black-box systems and need sufficient computational power and training data. [2]

Keyframe and visual feature extraction: Several video summarization approaches reveal the visual information with a representative key frame selection using clustering algorithm or similarity measures. While these methods try to highlight key visual scenes of as movie, they ignore semantic importance in the absence of text. [4]

Engaged Video Insights: Emerging work indicates that factors like watch time, retention curves, and interaction spikes from viewers offer us some complementary information on the significance of particular video segments. Yet few state-of-the-art summarization systems incorporate such behavioral signals into their ranking processes. [5]

Hybrid Summarization Methods: Hybrid methods try to leverage the benefits of extractive summarization along with generative models to offer better readability and coherence. Extractive approaches first find relevant information, while generative models directly generate a more natural summary representation. However, in practical terms a rich representation would measure how much work each modality does to tell

an informative and engaging story (in addition to text features) but few of such advanced systems include effective multimodal-fusion mechanisms that would dig into the details of engagement signals. [6]

Multimodal Fusion Approaches: There have been studies in video analytics recently focused on integrating more than one single piece of information, for example, transcripts, other visual cues and user interaction data. Combining the semantic and behavioural information in fusion-based approaches can give more informative summaries, however; currently this is quite restrictive for video summarization systems. [7]

2.2 COMPARISON OF PREVIOUS APPROACHES

Table 2.1: Comparison of Existing Video Summarization Approaches

Approach Type	Techniques Used	Advantages	Limitations
Visual-Based	Scene detection, keyframe extraction	Identifies important visual segments	Cannot understand semantic meaning
Text-Based	TF-IDF, TextRank, LSA	Captures textual meaning and context	Ignores visual and engagement aspects
Machine Learning / Deep Learning	CNN, RNN, Neural Networks	High accuracy, automated learning	Requires large datasets and high computation
Multimodal	Combination of text, audio, video	More comprehensive summaries	Complex integration, may miss engagement signals

Engagement-Based	Viewer retention, likes, comments	Identifies popular or engaging segments	May not reflect actual content importance

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

In the current scenario, video summarization is mainly performed using traditional methods or basic automated tools. Most users rely on manually watching videos or skimming through transcripts to extract important information. This process is time-consuming and inefficient, especially for long videos such as lectures, tutorials, and technical presentations.

Existing automated video summarization systems can be broadly categorized into visual-based and text-based approaches. Visual-based systems focus on extracting keyframes by detecting scene changes, motion patterns, or frame differences. While these methods help identify visually important segments, they fail to capture the actual meaning and context conveyed through speech.

On the other hand, text-based systems use transcripts or subtitles to generate summaries using Natural Language Processing (NLP) techniques such as TF-IDF and TextRank. These systems can understand the semantic content of the video but often ignore visual highlights and audience engagement patterns.

Some advanced systems use machine learning or deep learning techniques to improve summarization accuracy. However, these systems require large datasets, high computational power, and complex model training, making them less practical for lightweight or real-time applications.

Additionally, most existing systems do not provide features such as:

- Timestamp-based navigation
- Integration of engagement metrics (viewer retention, interaction peaks)
- Combination of multiple data sources (text, visual, and engagement)

  As a result, current video summarization approaches are limited in their ability to produce accurate, meaningful, and user-friendly summaries.

  Disadvantages of Existing System

- Time-consuming for users to watch entire videos
- Lack of semantic understanding in visual-based methods
- Ignoring visual and engagement features in text-based methods
- High computational cost in deep learning approaches

- No proper integration of multiple data sources

3.2 PROPOSED SYSTEM

The proposed system is a **Fusion-Based Video Summarization Framework** designed to generate concise, context-aware, and engagement-driven summaries from long-form video content. Unlike traditional summarization approaches that rely only on textual features, the system integrates transcript analysis with viewer engagement signals to enhance the relevance and accuracy of generated summaries.

The system collects video data including transcripts, timestamps, and engagement-related information such as retention patterns, interaction peaks, and sentiment intensity. Based on this data, it computes importance indicators by analyzing both semantic relationships between sentences and audience interaction behavior to identify key segments within the video.

Using these features, the fusion-based ranking mechanism dynamically combines textual importance scores with engagement metrics to prioritize high-impact content. The system further refines the selected segments using generative artificial intelligence to produce coherent and readable summaries, ensuring both accuracy and linguistic quality.

Additionally, an interactive interface provides visualization of summaries along with engagement insights such as retention curves and highlighted key segments. This integrated approach transforms conventional video summarization into an intelligent, user-centric, and engagement-aware content analysis system that improves accessibility and reduces information overload.

Advantages of Proposed System

- Fusion of transcript analysis with viewer engagement signals for improved summary relevance
- Engagement-aware ranking using retention patterns, interaction peaks, and sentiment analysis
- Temporal alignment of engagement data with transcript segments for precise highlight detection

3.3 FEASIBILITY STUDY

Feasibility study is an important step in system development that evaluates whether the proposed system is practical and achievable. It helps in analyzing the viability of the system from different perspectives such as technical, economic, and operational aspects.

1. Technical Feasibility

The proposed system is technically feasible as it uses widely available technologies and tools such as Python, Natural Language Processing (NLP), and machine learning libraries. The system utilizes APIs like the YouTube Transcript API and generative AI models for summarization, which are easily accessible.

The required tools such as Streamlit, Scikit-learn, and NLTK are open-source and can be implemented on standard computer systems without the need for high-end hardware. Therefore, the system can be developed and deployed efficiently with the available technical resources.

2. Economic Feasibility

The proposed system is cost-effective as it mainly relies on open-source tools and libraries. No expensive hardware or software is required for development and deployment.

Most of the technologies used, such as Python libraries and Streamlit, are free to use. Although APIs like generative AI services may involve minimal costs, they are manageable and can be optimized. Hence, the overall development and maintenance cost of the system is low.

3. Operational Feasibility

The system is user-friendly and easy to operate. It provides a simple web interface where users can enter a YouTube video URL and obtain summarized results.

No technical expertise is required to use the system, making it suitable for students, researchers, and general users. The system also provides structured outputs such as summaries, keyframes, and timestamps, improving usability and user experience.

4**. Time Feasibility**

The development of the proposed system can be completed within a reasonable time frame using existing tools and technologies. Since the system uses pre-built libraries and APIs, the implementation process is faster and more efficient.

The system also generates summaries within a short processing time, making it suitable for real-time or near real-time applications.

3.4 REQUIREMENTS

3.4.1 Hardware Requirements

The proposed system is a web-based application and does not require high-end hardware. It can run efficiently on a standard computer system.

Minimum Requirements:

- Processor: Intel Core i3 or equivalent
- RAM: 4 GB
- Storage: 20 GB free space
- Internet Connection: Required for API access

Recommended Requirements:

- Processor: Intel Core i5 or higher
- RAM: 8 GB
- Storage: 50 GB free space

3.4.2 Software Requirements

The system is developed using modern web technologies and machine learning frameworks to support transcript processing, engagement analysis, and AI-based summarization. The required software components include:

• Operating System: Windows / Linux / macOS

• Programming Language: Python

• Frontend Framework: Streamlit (for web-based interface)

• Machine Learning Libraries: Scikit-learn, NumPy, Pandas

• Natural Language Processing: NLTK / SpaCy

• Runtime Environment: Python 3.x

• Package Manager: pip

• Development Environment: Jupyter Notebook / Visual Studio Code

• Web Browser: Google Chrome / Mozilla Firefox

• Version Control System: Git

• APIs Used: YouTube Transcript API (for transcript extraction)

• AI Integration: Generative AI model (for abstractive summarization)

3.4.3 Functional Requirements

The system shall:

• Allow users to input a video URL for processing and analysis

• Retrieve and extract transcripts from video content using transcript APIs

• Preprocess textual data using NLP techniques such as tokenization and stop-word removal

• Convert transcript data into structured sentence-level representations

• Compute sentence importance using graph-based ranking techniques such as the TextRank algorithm

• Extract and analyze viewer engagement signals including retention patterns, interaction peaks, and sentiment intensity

• Align engagement signals with transcript segments using timestamp mapping

• Perform feature extraction and weighting to identify important content segments

• Apply a fusion-based scoring mechanism to combine textual and engagement features

• Generate extractive summaries based on ranked sentence importance

• Enhance summaries using generative artificial intelligence for improved readability

• Provide visualization of engagement insights such as retention graphs and interaction trends

3.4.4 Non-Functional Requirements

• Performance: The system shall generate video summaries within acceptable response time by efficiently processing transcripts and engagement data.

• Usability: The interface shall be intuitive, interactive, and user-friendly, allowing users to easily input video URLs and understand summarized outputs.

• Reliability: The system shall provide consistent and accurate summaries without failure during repeated usage and varying input conditions.

• Scalability: The architecture shall support handling multiple video inputs and enable future integration of multimodal features such as audio and visual analysis.

• Maintainability: The system shall follow a modular and well-structured design to facilitate easy updates, debugging, and extension of components.

• Compatibility: The application shall function properly across modern web browsers such as Google Chrome and Mozilla Firefox.

• Efficiency: The system shall optimize computational resources by using feature reduction techniques such as Principal Component Analysis.

CHAPTER 4

SOFTWARE DESIGN

4.1 ARCHITECTURE DESIGN

The architecture of the proposed Fusion-Based Video Summarization System is designed using a modular layered structure to ensure scalability, efficient processing, and seamless integration of multiple analytical components. The system is divided into interconnected modules including User Input Module, Transcript Extraction Module, Text Preprocessing Module, Summarization Module, Engagement Analysis Module, Fusion Model Module, Keyframe Extraction Module, Generative AI Module, and Output & Visualization Module.

The User Input Module allows users to provide a video URL for analysis. The Transcript Extraction Module retrieves transcript data using APIs, which is then processed by the Text Preprocessing Module through operations such as tokenization and stop-word removal to prepare clean textual data. The Summarization Module applies the TextRank algorithm to identify and rank important sentences from the transcript.

The Engagement Analysis Module evaluates viewer behaviour such as watch time, interaction peaks, and sentiment signals, which are then combined with textual importance in the Fusion Model Module to generate a unified relevance score for each segment. The Keyframe Extraction Module identifies important visual frames corresponding to high-impact segments. The Generative AI Module further refines the extracted content into coherent and readable summaries.

Finally, the Output Module presents the generated summaries, keyframes, and timestamps, while the Visualization Interface (Streamlit-based UI) displays results in an interactive and user-friendly manner, enabling users to quickly understand key insights from long-form video content.

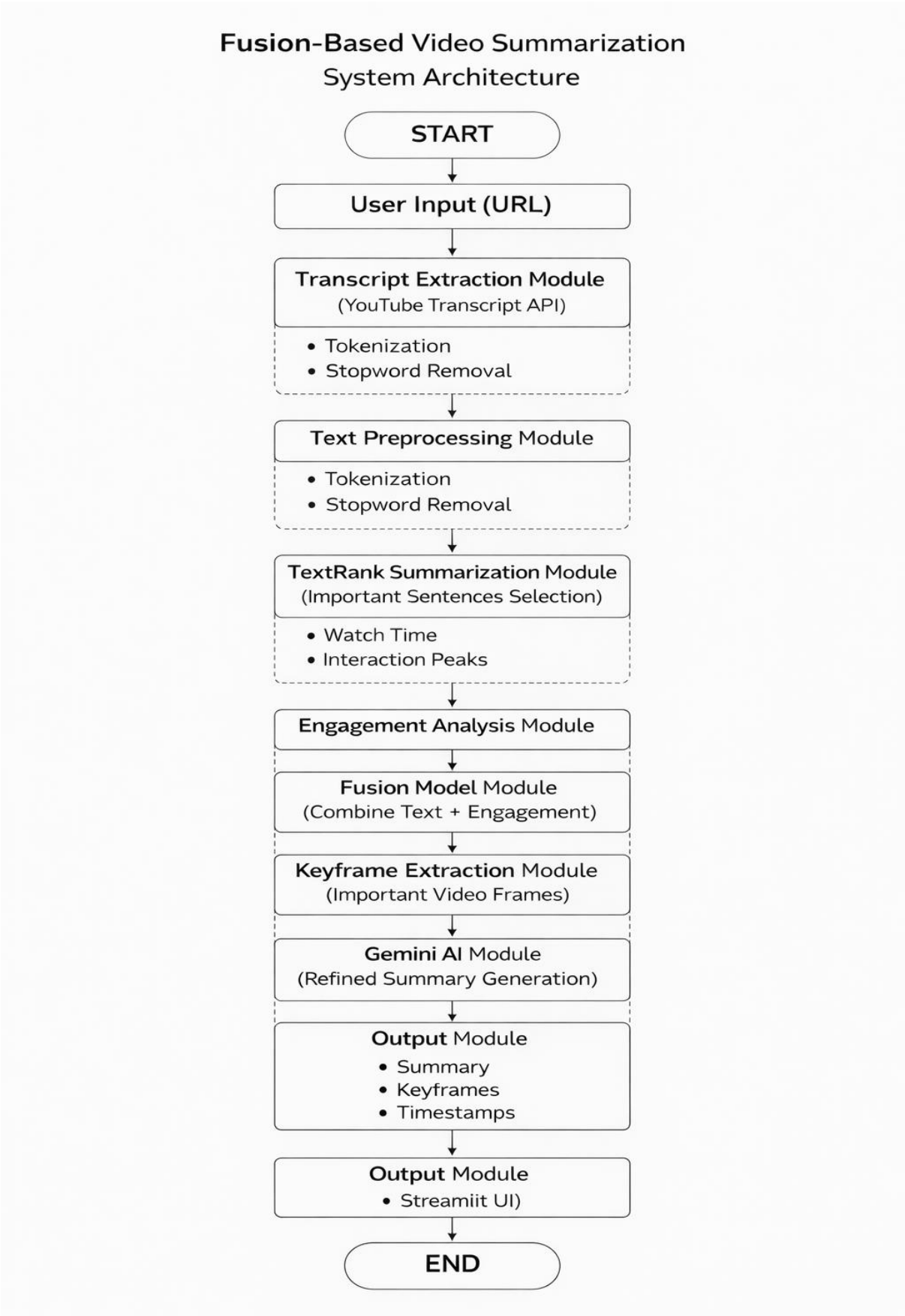

Fig 4.1: Fusion Based Video Summarization System Architecture

4.2 UML DIAGRAMS

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of the proposed system. These diagrams help in understanding how different components interact with each other and how the system processes user input to generate summarized output.

In this project, various UML diagrams such as Class Diagram, Sequence Diagram, Collaboration Diagram, Object Diagram, Use Case Diagram, and Activity Diagram are used to describe the system design. These diagrams provide a clear representation of system architecture, data flow, and interactions between different modules.

4.2.1 Class Diagram

Objective:

Define the static structure of the Fusion-Based Video Summarization system, including classes, their attributes, methods, and the relationships between system components.

Description:

Classes: VideoSummarizer, TranscriptService, TextProcessor, FeatureExtractor, TextRankSummarizer, EngagementAnalyzer, GeminiClient, StreamlitUI

VideoSummarizer – Core controller class that manages the complete summarization pipeline.

Attributes: apiKey, videoUrl, transcript, summary, engagementScore

Methods: extractTranscript(), preprocessText(), generateSummary(), analyzeEngagement()

TranscriptService – Responsible for fetching transcript data from external sources.

Attributes: apiUrl

Methods: fetchTranscript()

TextProcessor – Handles preprocessing of transcript text.

Attributes: tokens, cleanedText

Methods: tokenize(), removeStopwords(), preprocessText()

FeatureExtractor – Extracts important textual features for ranking.

Attributes: tfidfMatrix

Methods: computeTFIDF()

TextRankSummarizer – Generates extractive summaries using graph-based ranking.

Attributes: rankScores

Methods: computeSimilarity(), generateSummary()

EngagementAnalyzer – Processes viewer interaction and engagement signals.

Attributes: retentionScore, interactionScore

Methods: analyzeEngagement()

GeminiClient – Handles generative AI-based summary refinement.

Attributes: apiKey

Methods: generateAISummary()

StreamlitUI – Provides user interface for interaction and visualization.

Attributes: pageTitle, layout

Methods: setPageConfig(), displayInput(), displaySummary(), displayHighlights()

Relationships:

• VideoSummarizer coordinates and integrates all modules in the system pipeline.

• TranscriptService provides transcript data to VideoSummarizer.

• TextProcessor preprocesses transcript data before feature extraction.

• FeatureExtractor supplies features to TextRankSummarizer for ranking sentences.

• EngagementAnalyzer provides engagement metrics to enhance summary relevance.

• TextRankSummarizer generates extractive summaries used by GeminiClient.

• GeminiClient refines summaries into more coherent outputs.

• StreamlitUI displays summaries, highlights, and engagement insights to users.

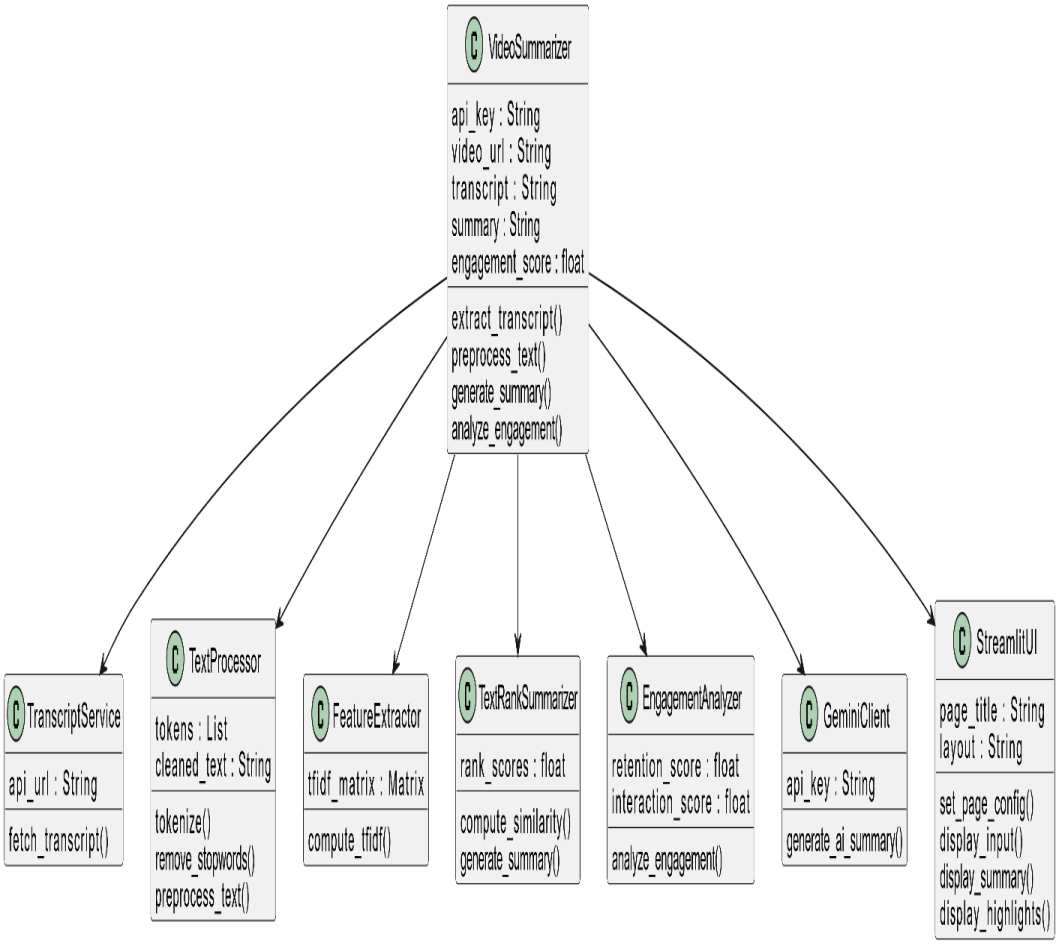

Fig 4.2.1. Class Diagram

4.2.2 Interaction Diagrams

Interaction diagrams describe how different components of the system communicate with each other. These diagrams show the sequence of operations and the flow of messages between objects.

4.2.2.1 Sequence Diagram

Objective:

Show how components interact in a sequence to generate a YouTube video summary and highlights.

Actors:

User, Web Interface, Transcript API, Text Processing Module, Summarization Engine, Gemini AI

Sequence:

1. User enters a YouTube URL into the system.
2. Web Interface sends a request to the Transcript API to fetch the video transcript.
3. Transcript API retrieves and returns the transcript back to the Web Interface.
4. Web Interface forwards the transcript to the Text Processing Module.
5. Text Processing Module performs tokenization and cleaning and applies TF-IDF and TextRank for importance scoring.
6. Processed data is sent to the Summarization Engine.
7. Summarization Engine performs engagement analysis and generates extractive summary.
8. The extractive summary is sent to Gemini AI for enhancement.
9. Gemini AI produces the final AI-generated summary.
10. The final summary is returned to the Web Interface.
11. Web Interface displays summary and key highlights.
12. User views the summarized content.

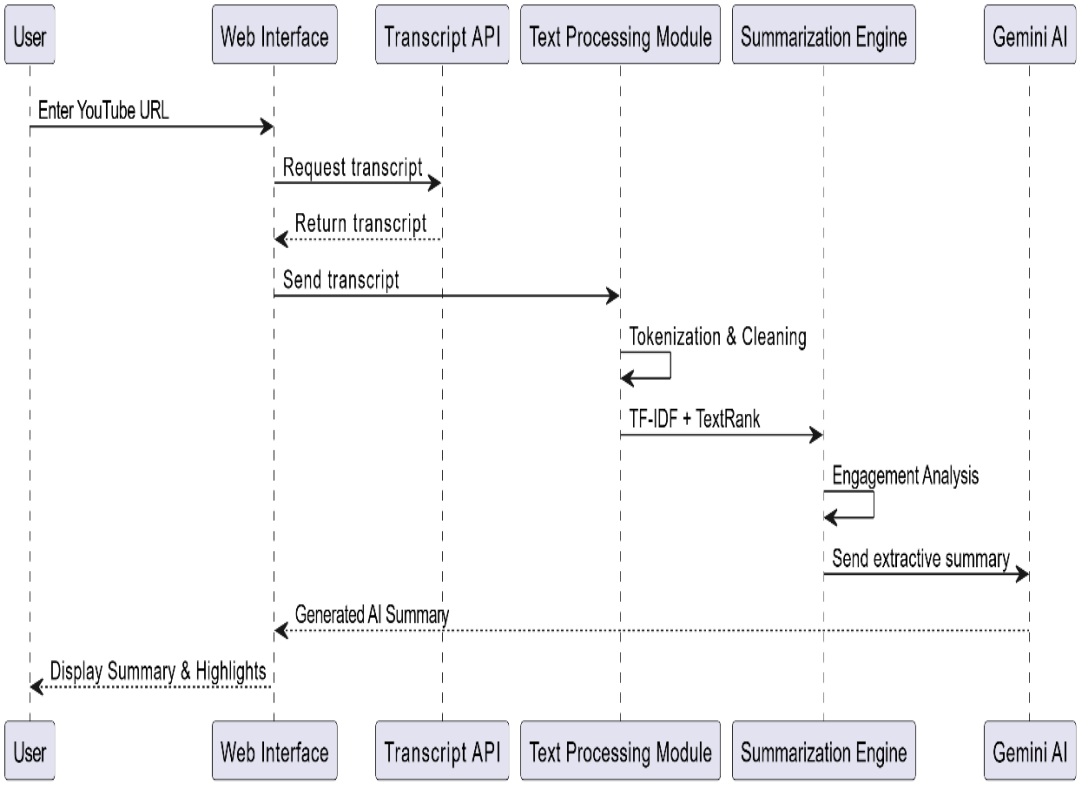

Fig.4.2.2.1 Sequence Diagram

4.2.2.2 Collaboration Diagram

The Collaboration Diagram shows how system components interact with each other to complete the summarization task. It focuses on the relationships between objects rather than the sequence of events.

In this system, the user interacts with the UI, which communicates with the VideoProcessor to extract transcripts. The processed text is passed through various modules such as TextProcessor, TextRank, and GeminiAI to generate the final summary.

This diagram highlights the cooperation between different modules in achieving the system's objective.

Explanation:

The collaboration diagram shows how system components interact and communicate with each other.

Key Components:

- User
- Web Interface
- Summarizer
- Engagement Module
- AI Module

Interactions:

- User sends request
- Interface forwards to backend
- Backend processes data
- AI module refines output

Difference from Sequence Diagram:

- Sequence diagram focuses on time
- Collaboration diagram focuses on structure

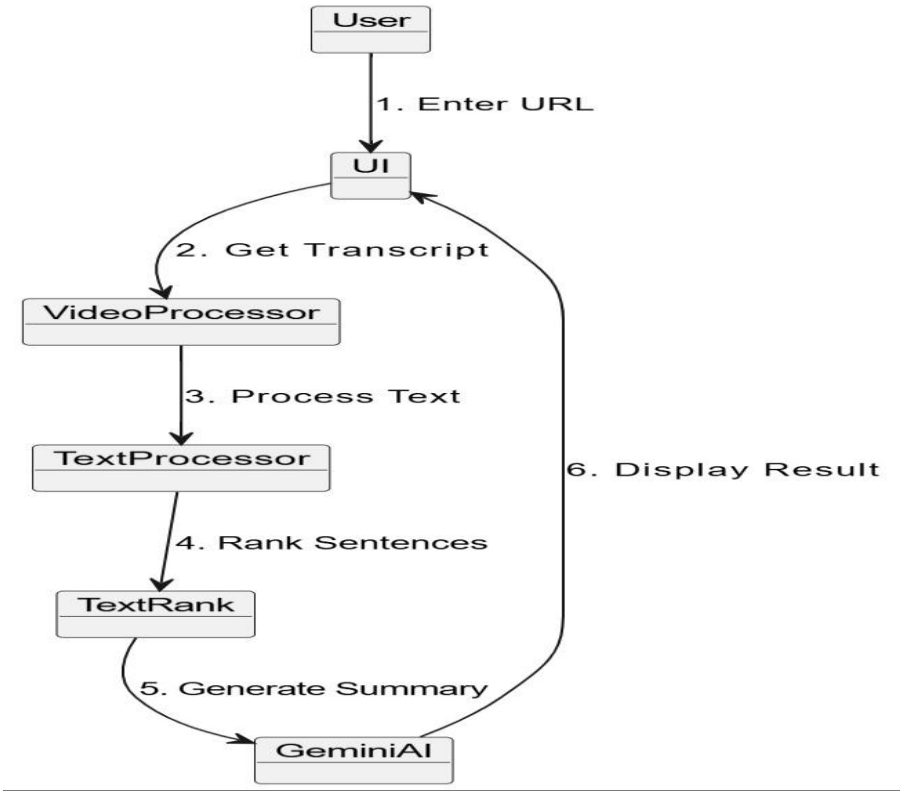

Fig. 4.2.2.2 Collaboration Diagram

4.2.3 Object Diagram

The Object Diagram represents a snapshot of the system at a particular moment in time. It shows instances of classes and their relationships.

In this project, objects such as user instance, UI instance, video instance, transcript object, and summary object are represented. The diagram illustrates how data flows from user input to the final summary output.

This helps in understanding how real-time data is handled within the system.

Explanation:

The object diagram represents real-time instances of classes and their relationships.

Example Instances:

- user1: User

- video1: Video
- transcript1: Transcript
- summary1: Summary

Key Points:

- Shows actual data values
- Represents system at a specific moment

Importance:

- Helps in debugging and testing
- Provides real-time system view

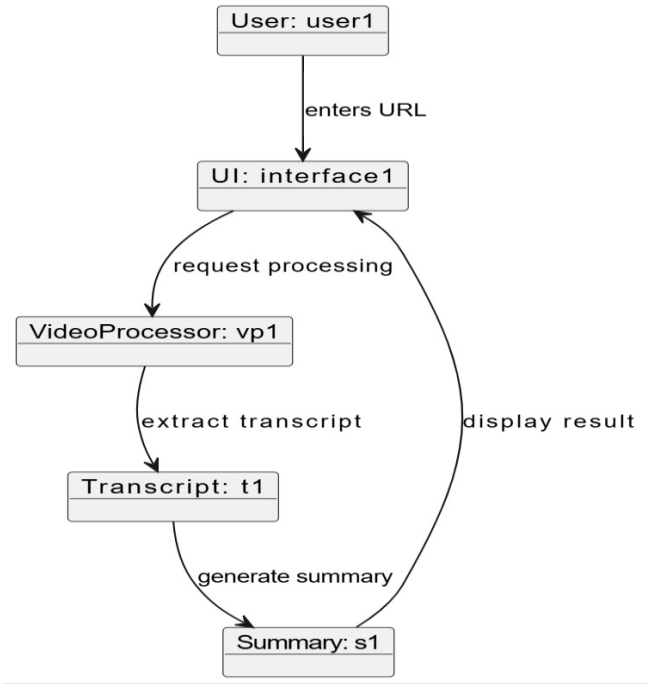

Fig. 4.2.3 Object Diagram

4.2.4UseCaseDiagram

The Use Case Diagram represents the interaction between the user and the system. It shows the different functionalities that the system provides to the user.

The main use cases in this project include:

• Enter YouTube video URL
• Extract transcript
• Generate summary
• View summarized output

This diagram helps in understanding the system from the user's perspective and identifies the main functionalities offered by the application.

Explanation:

The use case diagram describes how users interact with the system.

Actor:

- User

Use Cases:

- Enter video URL
- Extract transcript
- Generate summary
- View keyframes
- Analyze engagement

Relationships:

- Include relationship between summary and preprocessing
- Extend relationship for AI enhancement

Importance:

- Defines system functionality
- Helps in requirement analysis

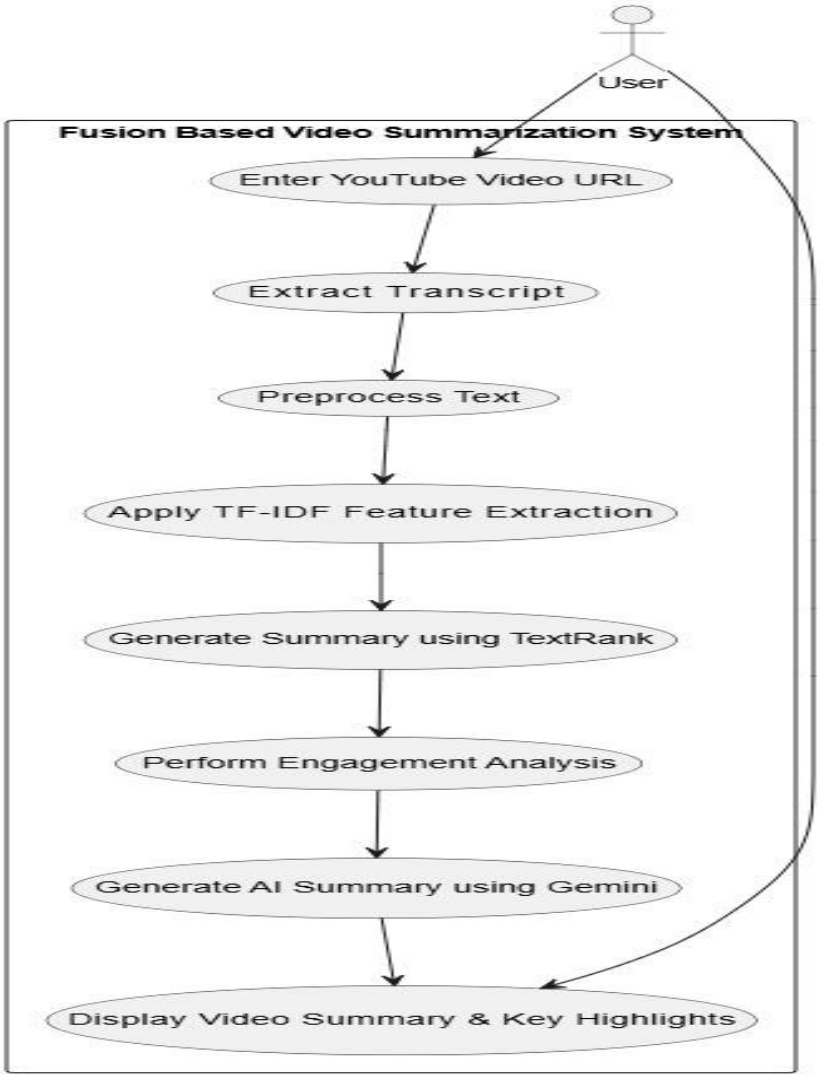

Fig. 4.2.4 Use Case Diagram

4.2.5ActivityDiagram

The Activity Diagram represents the workflow of the system. It shows the sequence of activities involved in the summarization process.

The process begins with user input, followed by transcript extraction, text preprocessing, feature extraction, summarization using TextRank, engagement analysis, and summary generation using Gemini AI. The final output is then displayed to the user.

This diagram provides a clear view of the system workflow and helps in understanding the step-by-step execution of tasks.

Explanation:

The activity diagram represents the workflow of the system.

Flow:

Start→InputURL→ExtractTranscript→Decision: Transcript available?

- Yes → Preprocess → Summarize → Output
- No → Show Error

Components:

- Initial node (start)
- Actions (processing steps)
- Decision nodes
- Final node (end)

Importance:

- Shows complete process flow
- Helps in algorithm understanding

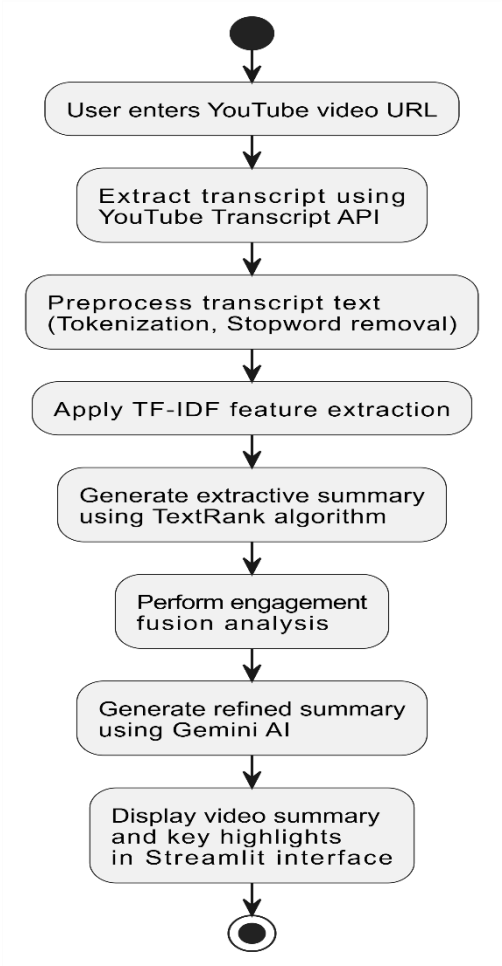

Fig. 4.2.5 Activity Diagram

4.2.6 Control Flow Diagram

The Control Flow Diagram illustrates the logical flow of the system, including decision-making processes. It shows how the system handles different conditions such as valid and invalid inputs.

When a user enters a YouTube video URL, the system first checks whether the URL is valid. If the URL is valid, the system proceeds with transcript extraction and summarization. If the URL is invalid, an error message is displayed to the user.

This diagram helps in understanding the control logic and decision-making process within the system.

Explanation:

The control flow diagram represents logical flow and decision-making in the system.

Key Steps:

- Input validation

- Condition checking
- Looping through sentences
- Ranking logic
- Output generation

Example Decisions:

- If transcript exists
- If summary length sufficient
- If API response valid

Importance:

- Helps in understanding logic
- Useful for debugging

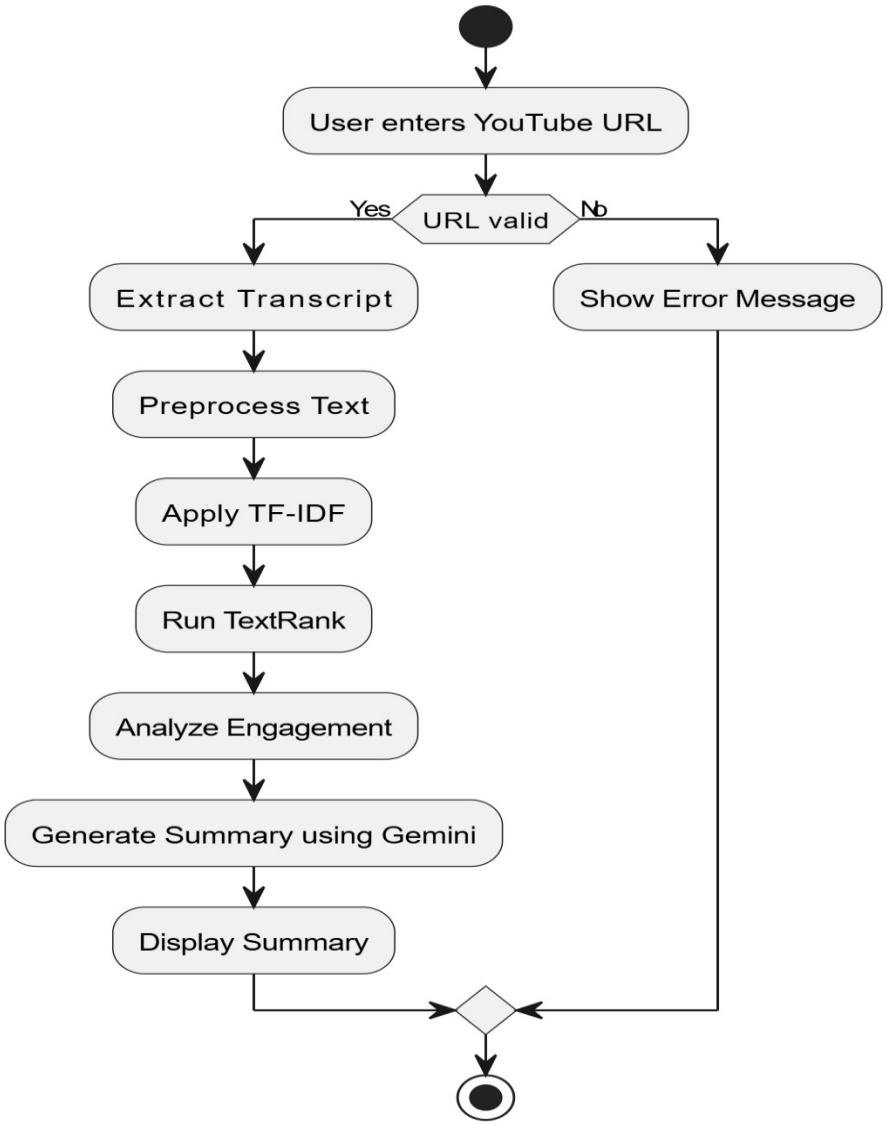

Fig. 4.2.6 Control Flow Diagram

4.3 DATABASE DESIGN

Database design defines how data is structured, stored, and managed within the system. In the proposed **Fusion-Based Video Summarization System**, a lightweight database is used to store user information, video details, and summarization history.

Although the system primarily processes data in real-time, a database is useful for maintaining user sessions, storing previously summarized videos, and improving user experience.

Main Entities (Tables)

1. User Table

Stores user account details.

Attributes:

- User_ID (Primary Key)
- Username
- Email
- Password (hashed)

2. Video Table

Stores information about the YouTube videos processed.

Attributes:

- Video_ID (Primary Key)
- Video_URL
- Video_Title
- Duration

3. Summary Table

Stores generated summaries and related data.

Attributes:

- Summary_ID (Primary Key)
- User_ID (Foreign Key)
- Video_ID (Foreign Key)
- Transcript_Text
- Extractive_Summary
- Abstractive_Summary
- Keyframes (timestamps)

4. User History Table (Optional)

Stores previously generated summaries for each user.

Attributes:

- History_ID
- User_ID
- Video_URL
- Summary
- Date_Time

Relationships

- One **User** can summarize multiple videos
- One **Video** can have multiple summaries
- Each **Summary** is linked to one user and one video

Key Features of Database Design

- **Efficient Storage:** Stores only essential data
- **Scalability:** Can handle multiple users and videos
- **Security:** Passwords stored in encrypted form
- **Data Retrieval:** Quick access to previous summaries
- **Modular Design:** Easy to extend in future

4.4 ALGORITHM / DESIGN METHODOLOGY

The YouTube Video Summarization framework is an integrated system that combines transcript-based natural language processing and user-engagement data that yields insightful summaries of long videos. This system integrates modules for transcript extraction, text preprocessing, graph-based summarization, engagement-driven ranking, and generative AI. By blending the importance of the text and the interaction signals of the audience, the framework identifies the most important text segments as well as the informative segments from the video. The key results turned into a special interface from which you can learn about the content you analyzed.

4.1 Video Input and Transcript Retrieval.

System starts with user interaction where the user gives a URL of YouTube video as an input through web interface. The YouTube Transcript API retrieves the relevant transcript consisting of time-oriented text segments that refer to the respective dialogue of the video. By maintaining the temporal order of the video, these segments allow the system to assess textual information and time-based engagement patterns.

The obtained transcript will be stitched together to obtain a continuous text document of the spoken video. The main document that will be used in further text processing and summarization tasks.

4.2 Text Preprocessing and Feature Preparation.

The textual data that are produced after the extraction of transcripts are preprocessed for uniformity and improved interpretation. The sentence segmentation splits the transcript into individual sentences. Tokenization is the process of splitting sentences into separate words, while removing stop words helps in reducing noise in the data. The techniques of text normalization such as lower casing and punctuation removal are done.

4.3 Extractive Summarization Using TextRank

The system does extractive summarization based on TextRank. The idea is that each sentence in the transcript will act as a node of a graph. TF-IDF vector representations and Cosine Similarity metrics are the tools used to measure similarities among different sentences.

Cosine similarity between two sentence vectors AAA and BBB, as:

$$Sim(A, B) = \frac{(A \cdot B)}{(||A|| \times ||B||)}$$

Where A and B refer to the TF-IDF vectors of two sentences.

The TextRank algorithm utilizes a mechanism akin to PageRank to assign importance scores to sentences after the construction of the similarity graph:

$$S(V_i) = (1 - d) + d \sum_{\{j \in In(V_i)\}} \left(\frac{w_{\{ji\}}}{\Sigma_{\{k \in Out(V_j)\}} w_{\{jk\}}} \right) S(V_j)$$

From above equation:

$S(V_i)$ is sentence i importance score.

d is damping factor (generally 0.85).

w_{ji} shows sentences similarity.

The extractive summary is made up of sentences sharing the highest ranking score.

4.4 Engagement Signal Analysis.

To make summarized segments more relevant, engagement indicators are incorporated. Viewer retention, peaks on interactions, and strength of sentiment from transcripts are part of these signals. Engagement analysis identifies the segments that attract more attention or emotion from viewers.

By assessing these actions, the system detects portions of the video that are crucial according to the viewer's perspective.

4.5 Engagement Fusion Ranking Model.

To merge the significance of the texts with the information on engagement, we propose a Participation Fusion Ranking Model. This model measures a video's importance using text rank scores in conjunction with engagement metrics. These metrics include viewer retention over time, interaction peaks and sentiment magnitude of comments.

The fusion score of each sentence is determined as:

$$F_i = \alpha TR_i + \beta R_i + \gamma I_i + \delta S_i$$

4.6 Generative AI-Based Abstractive Summarization.

The framework incorporates the AI techniques of generative Artificial Intelligence that can generate the abstractive summary for better readability and coherence. Distinction

Between Generative and Extractive Models: Unlike approaches that pick sentences, a generative model generates sentences capturing the essential meaning of the original transcript. The summaries are a concise and a natural representation of the key ideas in the video.

4.7 Dimensionality Reduction.

To enhance computing efficiency, dimensionality reduction on textual feature vectors is applied. We perform Principal Component Analysis (PCA) on TF-IDF vectors to reduce their dimensions while preserving most of the significant variance present in the data.

The process minimizes repetition in sentence similarity representation, allowing for faster similarity calculations.

4.8 Interactive Visualization Interface.

The final results are summarized using an interactive web app built on Streamlit. The tool allows users to submit video URLs and access a summarized version of the video along with its engagement metrics. The interactive design helps users quickly understand the highlights of the long-form video content.

4.9 Summary Report Generation.

The system's final output will contain an extractive summary, a generative summary along with engagement insights. You can quickly get a sense of the main ideas from the video without following the entire thing, thanks to the video highlights report. This method allows you to consume and analyze content efficiently.

CHAPTER 5
IMPLEMENTATION

5.1 TECHNOLOGIES USED

The proposed Fusion-Based Video Summarization System is developed using a combination of programming languages, frameworks, libraries, and APIs. These technologies enable efficient processing of video data, implementation of NLP techniques, and development of a user-friendly web interface.

1. Programming Language

Python

- Main language used for development
- Supports machine learning, NLP, and data processing
- Easy to implement and widely used in AI applications

2. Development Tools

- Visual Studio Code / PyCharm – Code development and debugging
- Git – Version control system for managing project code

3. Libraries and Frameworks

Streamlit

- Used to build the web interface
- Provides interactive UI for user input and output display

NLTK (Natural Language Toolkit)

- Used for text preprocessing
- Performs tokenization, stop-word removal, and sentence splitting

Scikit-learn

- Used for machine learning operations
- Implements:
- TF-IDF vectorization
- Cosine similarity

NumPy

- Used for numerical computations
- Helps in handling arrays and mathematical operations

Pandas

- Used for data manipulation and analysis

NetworkX

- Used to implement the TextRank algorithm
- Builds graph-based ranking system

4. APIs and External Services YouTube Transcript API

- Extracts transcripts from YouTube videos
- Provides text data for analysis

Google Gemini API

- Used for generative AI summarization
- Produces refined and human-readable summaries

5. Database (Optional)

PostgreSQL / Neon Database

- Stores user data and summarization history
- Supports authentication and session management

6. Operating Systems

- Windows
- Linux
- macOS

5.2 MODULES DESCRIPTION

Here's a **clear and slightly expanded elaboration** of each module, keeping it detailed but still suitable for a report:

5.2.1 User Interface and Task Management Module

This module serves as the entry point of the system and is implemented using Streamlit in the main application file. It provides an interactive interface where users can input a YouTube video URL and initiate the summarization process. The module manages the workflow by coordinating between backend components and ensuring smooth data flow. It also presents outputs such as generated summaries, extracted keyframes, and analytics in a structured and user-friendly manner, improving overall usability.

5.2.2 Transcript Extraction Module

This module is responsible for retrieving the textual content of the video using the YouTube Transcript API. It extracts both the transcript text and corresponding timestamps, which are essential for aligning summaries with specific video segments. The module ensures accurate and efficient data retrieval and handles cases where transcripts may be unavailable or incomplete.

5.2.3 Text Preprocessing Module

This module prepares the raw transcript for further analysis using Natural Language Processing techniques. It performs tokenization to break text into smaller units, removes stop-words to eliminate irrelevant terms, and applies sentence segmentation to structure the content. These preprocessing steps reduce noise, improve data quality, and enhance the effectiveness of subsequent analysis.

5.2.4 Feature Extraction Module

This module transforms processed textual data into numerical representations using TF-IDF (Term Frequency–Inverse Document Frequency) vectorization. It assigns importance scores to words based on their frequency and uniqueness within the transcript. Additionally, it computes similarity between sentences, which is crucial for identifying relationships and determining the significance of different parts of the content.

5.2.5 Extractive Summarization Module

This module implements the TextRank algorithm to generate an initial summary. It constructs a graph where sentences are represented as nodes and edges indicate similarity between them. Using a PageRank-based ranking mechanism, the module assigns importance scores to sentences. The highest-ranked sentences are selected to form a concise extractive summary that captures the core information of the video.

5.2.6 Engagement Fusion Module

This module enhances the summarization process by incorporating user engagement metrics such as viewer retention, sentiment intensity, and interaction levels (likes, comments, etc.). It combines these factors with TextRank scores to identify the most impactful segments of the video. This fusion approach ensures that the summary is not only textually important but also reflects audience interest and engagement.

5.2.7 Multimodal Fusion Module

This module integrates multiple data modalities, including textual content, visual features, and audio signals. By combining information from different sources, it provides a more comprehensive understanding of the video. This multimodal approach improves summary quality by capturing context, emotions, and visual significance that may not be evident from text alone.

5.2.8 Generative AI Module

This module uses the Gemini API to refine the extractive summary into a more coherent and human-readable format. It restructures sentences, improves readability, and enhances the overall presentation of the summary. This step bridges the gap between raw extractive output and polished natural language summaries.

5.2.9 Keyframe Extraction Module

This module identifies important timestamps based on summary and engagement scores and extracts corresponding video frames. These keyframes visually represent significant moments in the video, allowing users to quickly grasp key points and navigate to important segments efficiently.

5.2.10 Database and History Module

This module manages the storage of user inputs, video metadata, and generated summaries using a database system. It maintains a history of processed videos, enabling users to revisit previous results without reprocessing. This improves efficiency and enhances user experience by providing persistent access to past analyses.

5.2.11 Testing and Verification Module

This module ensures the reliability and correctness of the system through systematic testing. It includes scripts for validating API integrations, checking summarization accuracy, and verifying the proper functioning of each module. It also helps identify bugs, performance issues, and inconsistencies, ensuring the system operates as intended.

5.3 CODING

5.3.1 Project Folder Structure

The proposed Fusion-Based Video Summarization System is organized into multiple modules including backend processing, AI integration, testing, and frontend interface. The modular structure improves code readability, maintainability, and scalability.

The system is divided into backend, frontend, testing, and configuration components.

A. Backend

The backend module is responsible for processing video data, performing NLP operations, and generating summaries using AI techniques.

Main Files in Backend:

- app_enhanced_fixed.py
- engagement_fusion.py
- multimodal_fusion.py
- fix_nltk.py

Responsibilities:

- Extract transcripts from YouTube videos
- Perform text preprocessing using NLP
- Apply TF-IDF and TextRank algorithms
- Perform engagement-based scoring
- Integrate Generative AI models

B. Frontend

The frontend is developed using Streamlit and provides a simple and interactive interface for users.

Main Features:

- Accept YouTube video URL
- Display transcript
- Show extractive and abstractive summaries
- Display keyframes and timestamps
- Visualize engagement analytics

C. Testing Module

The testing module ensures the correctness and reliability of the system.

Files Included:

- test_gemini.py
- test_textrank_refinement.py
- verify_env.py

- verify_fusion.py
- verify_multimodal.py
- syntax_check.py

D. Configuration and Setup

- requirements_enhanced.txt – Contains dependencies
- install scripts – Used for environment setup

5.3.2 Backend Implementation

The backend is implemented using Python and consists of multiple components such as transcript extraction, preprocessing, summarization, engagement fusion, and AI integration.

File: app_enhanced_fixed.py

Purpose:

This is the main application file that integrates all backend functionalities and connects them with the frontend.

Key Functionalities:

- Accepts YouTube video URL
- Extracts transcript
- Applies preprocessing
- Performs TextRank summarization
- Applies engagement fusion
- Generates AI-based summary
- Displays output

Code: Transcript Extraction

```
from youtube_transcript_api import YouTubeTranscriptApi

def get_transcript(video_id):
    transcript = YouTubeTranscriptApi.get_transcript(video_id)
```

```
    full_text = " ".join([t['text'] for t in transcript])
    return full_text
```

Explanation:

This function extracts subtitles from YouTube videos and converts them into continuous text.

Code: Text Preprocessing

```
import nltk
from nltk.tokenize import sent_tokenize
from nltk.corpus import stopwords

def preprocess_text(text):
    sentences = sent_tokenize(text)
    clean_sentences = []

    for sentence in sentences:
        words = sentence.lower().split()
        words = [w for w in words if w not in stopwords.words('english')]
        clean_sentences.append(" ".join(words))

    return clean_sentences
```

Explanation:

This step removes unnecessary words and prepares the text for summarization.

Code: TF-IDF Vectorization

```
from sklearn.feature_extraction.text import TfidfVectorizer

def compute_tfidf(sentences):
    vectorizer = TfidfVectorizer()
    vectors = vectorizer.fit_transform(sentences)
    return vectors
```

Code: TextRank Algorithm

```
from sklearn.metrics.pairwise import cosine_similarity
import networkx as nx

def textrank_summary(sentences):
    vectors = compute_tfidf(sentences)
    similarity_matrix = cosine_similarity(vectors)

    graph = nx.from_numpy_array(similarity_matrix)
    scores = nx.pagerank(graph)

    ranked_sentences = sorted(((scores[i], s) for i, s in enumerate(sentences)),
reverse=True)
    summary = " ".join([s for _, s in ranked_sentences[:5]])

    return summary
```

Explanation:

This algorithm ranks sentences based on importance and selects the top ones for summary generation.

File: engagement_fusion.py

Purpose:

Enhances summarization by incorporating user engagement metrics**.**

Code: Engagement Fusion

```
def calculate_fusion_score(text_score, retention, sentiment, interaction):
    final_score = (0.2 * text_score) + (0.5 * retention) + (0.2 * sentiment) + (0.1 *
interaction)
    return final_score
```

Explanation:

This function combines multiple factors to improve the accuracy of summary selection.

File: multimodal_fusion.py

Purpose:

Combines text, visual, and engagement features for improved summarization.

Explanation:

This module integrates multiple data sources to generate more meaningful summaries.

Code: Generative AI Integration

```
import google.generativeai as genai

def generate_ai_summary(text):
    model = genai.GenerativeModel("gemini-flash-latest")
    response = model.generate_content(text)
    return response.text
```

Explanation:

This function generates a refined and human-readable summary using AI.

5.3.3 Frontend Implementation

The frontend is implemented using Streamlit and provides interaction between the user and the system.

User Input Interface

```
import streamlit as st

st.title("Fusion-Based Video Summarizer")

video_url = st.text_input("Enter YouTube Video URL")

if st.button("Generate Summary"):
    st.write("Processing video...")
```

Displaying Output

```
st.subheader("Generated Summary")
st.write(summary)
```

```
st.subheader("Keyframes")
st.write(keyframes)
```

Visualization (Optional Analytics)

```
import matplotlib.pyplot as plt

def plot_engagement(data):
    plt.plot(data)
    plt.xlabel("Time")
    plt.ylabel("Engagement")
    plt.title("Engagement Analysis")
    plt.show()
```

5.3.4 System Workflow in Code

- User inputs YouTube URL
- Transcript is extracted
- Text is preprocessed
- TF-IDF and TextRank applied
- Engagement fusion improves ranking
- Keyframes are extracted
- AI generates final summary
- Results displayed on UI

CHAPTER 6

TESTING

6.1 TEST CASES

Testing is an essential phase in software development used to verify that the system works correctly and meets the specified requirements. The proposed Fusion-Based Video Summarization System is tested using different testing methods to ensure accuracy, reliability, and performance.

Table 6.1: Test Cases of Video Summarization System

Test Case ID	**Description**	**Input**	**Expected Output**	**Actual Result**	**Status**
TC01	Valid YouTube URL	YouTube link	Transcript extracted successfully	Transcript displayed	Pass
TC02	Invalid URL	Wrong link	Error message	Error shown	Pass
TC03	No transcript video	Video without subtitles	Error message	Error displayed	Pass
TC04	TextRank summarization	Valid transcript	Extractive summary generated	Summary displayed	Pass
TC05	Gemini summarization	Valid transcript	Refined summary	Output generated	Pass
TC06	Engagement fusion	Valid video	Key highlights detected	Highlights shown	Pass

TC07	Multimodal fusion	Valid video	Combined summary	Output generated	Pass
TC08	User login	Valid credentials	Login successful	Login success	Pass

6.2 TESTING METHODS

Different types of testing methods are used to evaluate the system performance.

6.2.1 Unit Testing

Unit testing is performed to test individual modules of the system.

- Testing transcript extraction
- Testing TextRank algorithm
- Testing Gemini API integration
- Testing engagement fusion module

Ensures each component works independently

6.2.2 Integration Testing

Integration testing is used to verify that different modules work together correctly.

- Transcript extraction + preprocessing
- TextRank + engagement fusion
- Backend + frontend integration

Ensures smooth data flow between module

6.2.3 System Testing

System testing is performed on the complete system to verify overall functionality.

- End-to-end testing of video summarization
- UI testing using Streamlit
- Performance testing with different video lengths

Ensures system meets all requirements

6.3 RESULTS AND ANALYSIS

The testing results show that the system performs efficiently and accurately under different conditions.

- The system successfully extracts transcripts for most videos
- TextRank provides meaningful extractive summaries
- Engagement fusion improves highlight detection
- Gemini AI enhances readability and structure
- The system handles errors such as invalid URLs and missing transcripts

Overall, the system demonstrates good performance, accuracy, and usability.

CHAPTER 7

RESULTS&DISSCUSSIONS

This chapter presents the outputs of the proposed Fusion-Based Video Summarization System and analyzes its performance. The system was tested on different YouTube videos to evaluate its accuracy, efficiency, and usability.

7.1 Dashboard Overview

The dashboard interface provides a summary of the user's study activity, including task completion rate, total study time, upcoming tasks, and AI recommendations. The dashboard serves as the central interface where users can monitor progress and access different features of the study planner.

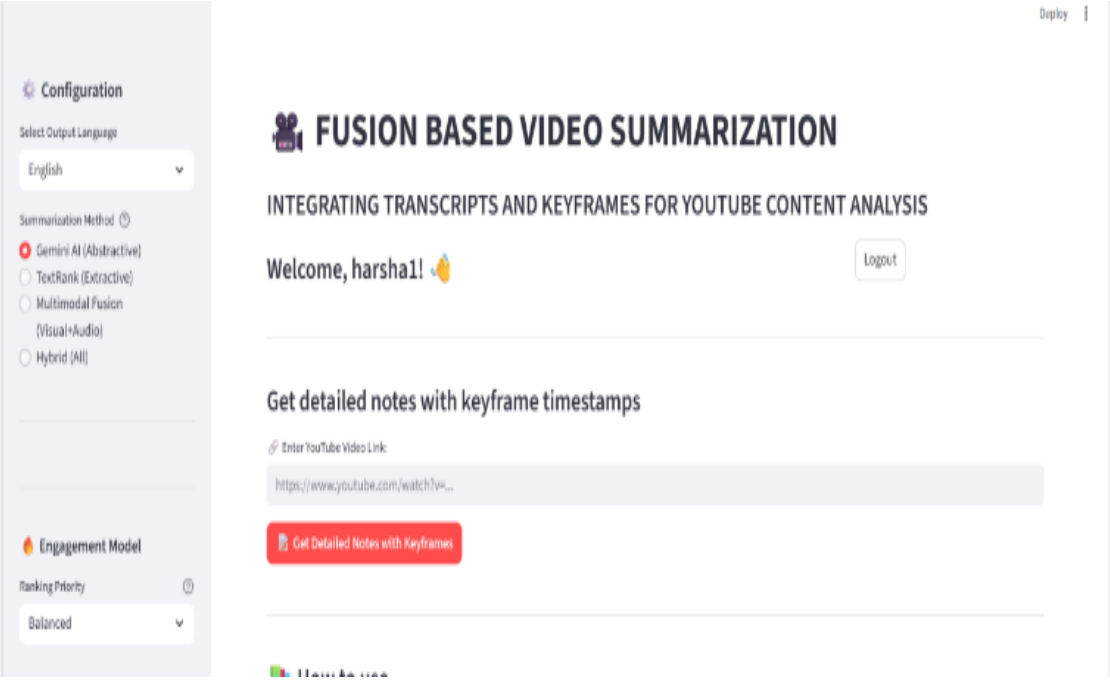

Figure 7.1: Dashboard Interface of the Fusion based video summarization

- Displays the main interface of the application
- Allows users to input YouTube video URL
- Provides options for selecting summarization methods

7.2 Video Input Interface

The Video Input Interface allows users to provide a YouTube video URL as input. This is the starting point of the system workflow.

The system validates the entered URL and checks whether the transcript is available. If the input is valid, the system proceeds with transcript extraction. If the URL is invalid, an error message is displayed.

This interface ensures smooth and efficient interaction between the user and the system.

7.3 AI-Based Summary Generation

This section displays the summary generated by the system using TextRank and Gemini AI.

Initially, the system generates an extractive summary using the TextRank algorithm. Then, the Gemini generative AI model refines the summary to improve readability and coherence.

The output is presented in a clear and structured format, making it easy for users to understand the main content of the video without watching it entirely.

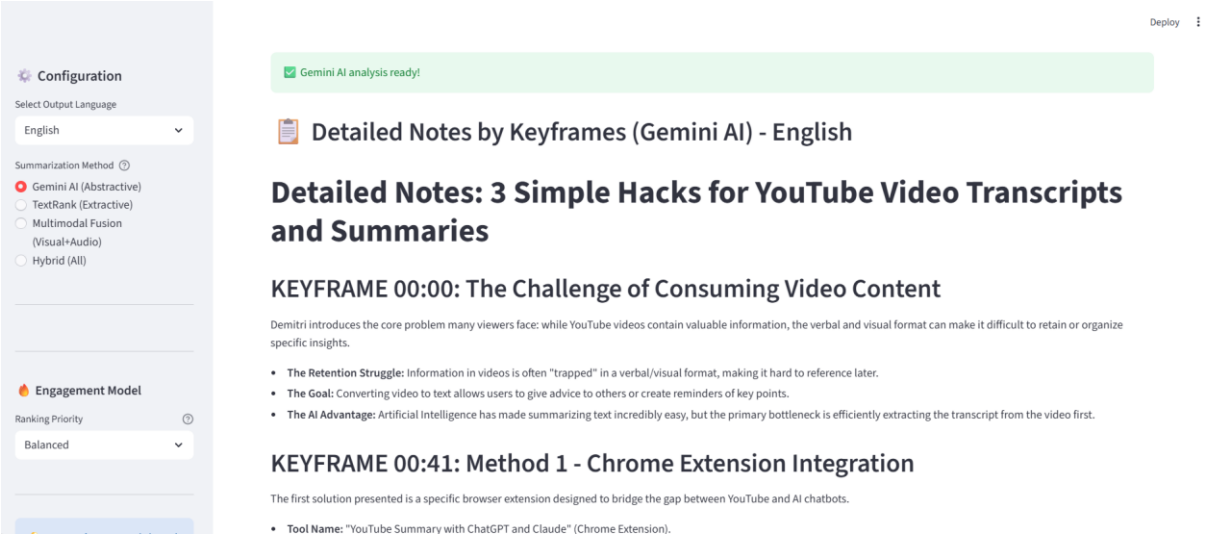

Figure 7.3: AI summary of the Fusion based video summarization

7.4 Key Frame / Highlight Visualization

The system also identifies important segments of the video using engagement signals and transcript analysis.

These key segments are highlighted to help users quickly locate important parts of the video. This improves user experience by allowing quick navigation to relevant sections. The visualization may include timestamps or highlighted text representing key moments in the video.

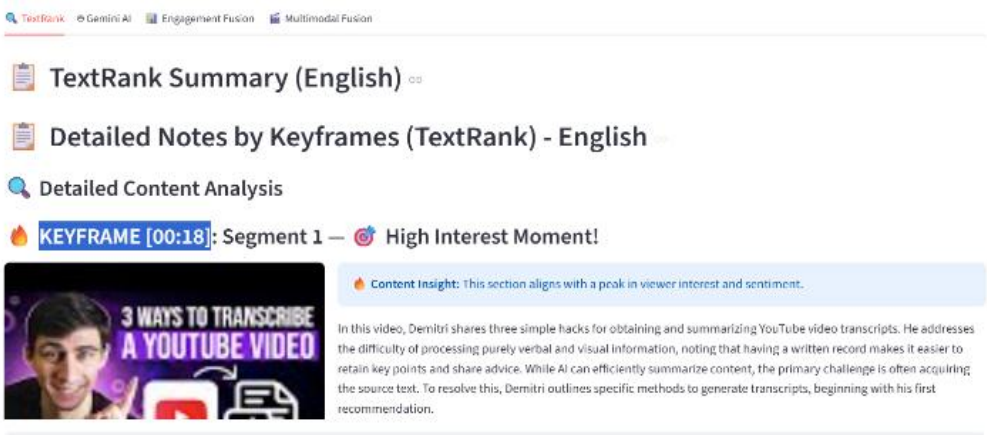

Figure 7.4: TextRank Summary With Engagement Fusion

7.5 Summary Analytics and Insights

The system provides additional insights into the generated summary. These include:

- Number of sentences in the transcript
- Number of sentences selected for the summary
- Processing time
- Summary length reduction percentage

These analytics help evaluate the efficiency of the summarization process and provide useful information to users.

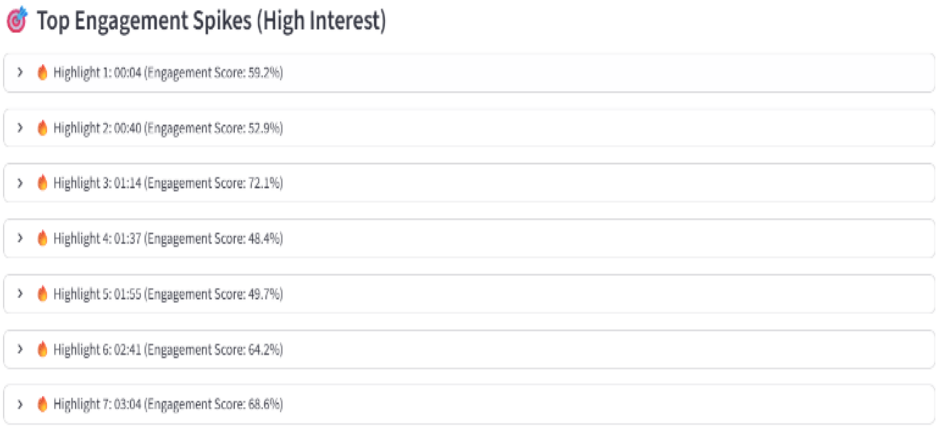

Figure 7.5: Engagements spikes

7.6 Explainable AI-Based Output

The system provides an explainable output by showing how the summary was generated. It highlights important sentences selected by the TextRank algorithm and shows how the generative AI refined them.

This improves transparency and helps users understand the reasoning behind the generated summary.

7.7 System Overall Performance

The overall performance of the system is evaluated based on accuracy, speed, and usability.

The system successfully generates summaries for long video transcripts within a short time. The combination of extractive summarization and generative AI improves both the quality and readability of the output.

The Streamlit interface ensures smooth interaction, and the system performs efficiently across different types of videos.

Metric	Traditional Methods	Proposed Fusion System
Summary Accuracy	Moderate – single modality	High – multimodal fusion
Engagement Awareness	Not considered	Incorporated via fusion engine
Readability	Often fragmented	Improved by generative AI
Timestamp Navigation	Not available	Provided with clickable links
Processing Efficiency	Variable	Optimized with PCA and TF-IDF

7.7 Comparsion of summarization technique

CHAPTER 8

CONCLUSION AND FUTURE WORK

8.1 SUMMARY OF THE PROJECT

The proposed Fusion-Based Video Summarization System integrating transcripts for YouTube content analysis successfully addresses the problem of extracting meaningful information from lengthy videos. The system combines Natural Language Processing techniques, TextRank algorithm, engagement analysis, and Generative AI to produce accurate and structured summaries.

The system takes a YouTube video as input, extracts its transcript, processes the text, and generates both extractive and abstractive summaries. It also identifies important timestamps and keyframes, enabling users to quickly navigate through the video content

8.2 ACHIEVEMENTS

The project has achieved the following objectives:

- Developed an automated system for video summarization
- Successfully integrated transcript-based NLP techniques
- Implemented TextRank algorithm for extractive summarization
- Applied engagement fusion to improve summary relevance
- Integrated Generative AI for human-readable summaries
- Designed a user-friendly interface using Streamlit
- Enabled keyframe extraction and timestamp navigation

8.3 LIMITATIONS

Despite its advantages, the system has some limitations:

- Depends on availability of YouTube transcripts

- Limited accuracy for videos with poor subtitles
- Engagement metrics are simulated or limited
- High dependency on external APIs (Gemini, YouTube API)
- Performance may vary for very long videos

8.4 FUTURE ENHANCEMENTS

The system can be improved further by implementing the following features:

- Integration of real-time video processing
- Advanced deep learning models (Transformers, BERT)
- Improved multimodal fusion using audio and visual signals
- Real-time engagement data integration
- Support for multiple languages
- Mobile application development
- Enhanced visualization dashboards

REFERENCES

[1] R. Mihalcea and P. Tarau, “TextRank: Bringing order into texts,” in *Proc. Conf. Empirical Methods in Natural Language Processing (EMNLP)*, Barcelona, Spain, 2004, pp. 404–411.

[2] J. Devlin, M. W. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proc. North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*, Minneapolis, MN, USA, 2019, pp. 4171–4186.

[3] A. Vaswani et al., “Attention is all you need,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[4] B. T. Truong and S. Venkatesh, “Video abstraction: A systematic review and classification,” *ACM Transactions on Multimedia Computing, Communications, and Applications*, vol. 3, no. 1, pp. 1–37, 2007.

[5] M. Otani, Y. Nakashima, E. Rahtu, and J. Heikkilä, “Rethinking the evaluation of video summaries,” in *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, Long Beach, CA, USA, 2019, pp. 7596–7604.[6] J. Zhang, W. Chao, F. Sha, and K. Grauman, “Video summarization with long short-term memory,” in *Proc. European Conf. Computer Vision (ECCV)*, Amsterdam, Netherlands, 2016, pp. 766–782.

[7] H. Zheng, Y. Liu, and L. Wang, “Multimodal video summarization using deep neural networks,” *IEEE Access*, vol. 8, pp. 184725–184736, 2020.

[8] A. G. Money and H. Agius, “Video summarisation: A conceptual framework and survey of the state of the art,” Journal of Visual Communication and Image Representation, vol. 19, no. 2, pp. 121–143, 2008.

[9] Y. Gong and X. Liu, “Video summarization using singular value decomposition,” in Proc. IEEE Conf. Computer Vision and Pattern Recognition, 2001, pp. 174–180.

[10] K. Zhang, W. Chao, F. Sha, and K. Grauman, “Summary transfer: Exemplar-based subset selection for video summarization,” in Proc. IEEE Conf. Computer Vision and Pattern Recognition, 2016, pp. 1059–1067.

[11] S. Narayan, S. B. Cohen, and M. Lapata, “Ranking sentences for extractive summarization with reinforcement learning,” in Proc. NAACL-HLT, 2018, pp. 1747–1759.

[12] J. Nallapati, B. Zhou, C. Gulcehre, and B. Xiang, “Abstractive text summarization using sequence-to-sequence RNNs and beyond,” in Proc. SIGNLL Conf. Computational Natural Language Learning, 2016.

[13] S. Fabbri, W. Li, T. She, S. Li, and D. Radev, “Multi-News: A large-scale multi-document summarization dataset and abstractive hierarchical model,” in Proc. ACL, 2019.

[14] Y. Liu and M. Lapata, “Text summarization with pretrained encoders,” in Proc. EMNLP, 2019.

[15] K. Cho et al., “Learning phrase representations using RNN encoder–decoder for statistical machine translation,” in Proc. EMNLP, 2014.